

BUREAU OF INDIAN STANDARDS

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भारतीय मानक मसौदा

**सर्जिकल उपकरण — स्टेनलेस स्टील सर्जिकल उपकरणों का
संक्षारण प्रतिरोध — परीक्षण पद्धति**

(IS 7531 का दूसरा पुनरीक्षण)

Draft Indian Standard

**Surgical Instruments — Corrosion Resistance of
Stainless Steel Surgical Instruments — Methods of
Tests**

(Second Revision of IS 7531)

ICS 11.040.30

Surgical Instruments Sectional Committee, MHD 01

Last date for comments: 30 September 2026

FOREWORD

(Formal clauses will be added later)

This Indian Standard (Second Revision) was first published in 1975 as ‘Method for boiling and autoclaving test for corrosion resistance of stainless steel surgical instruments’. This revision aligns the cross references to the latest standards, incorporates revised designations for steel and revised certification clause.

In reporting the result of a test or analysis made in accordance with this standard, is to be rounded off, it shall be done in accordance with IS 2 : 2022 ‘Rules for rounding off numerical values (second revision)’. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

1 SCOPE

This Indian Standard specifies the methods for testing of corrosion resistance of surgical instruments made from stainless steel.

2 REFERENCES

The standards given below contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

<i>IS No.</i>	<i>Title</i>
IS 1070 : 2023	Reagent Grade Water Specification (Fourth Revision)

3 TYPE OF TESTS

3.1 The following tests are specified for determining the corrosion resistance:

- a) Copper Sulphate Test
- b) Boiling Water Test
- c) Autoclaving Test

3.2 The purchaser shall specify whether all the tests are to be carried out, or if not, which of the tests are to be performed. If the purchaser does not so specify, it is left to the discretion of the manufacturer as to which test to apply.

4 COPPER SULPHATE TEST

4.1 Apparatus

Glass or ceramic beaker.

4.2 Reagents

4.2.1 Copper (II) Sulphate Pentahydrate

4.2.2 Sulphuric Acid (rd 1.84)

4.2.3 Distilled Water

Conforming to IS 1070.

4.3 Preparation of Test Solution

Dissolve 4 g of copper sulphate (**4.2.1**) in 10 g of sulphuric acid (**4.2.2**). Cautiously, add the above to 90 ml of distilled water (see **4.2.3**) taken in glass or ceramic beaker (**4.1**).

4.4 Preparation of Sample

Scrub the surgical instrument using soap and water, rinse thoroughly in distilled water, dip in 95 percent (v/v) ethanol and dry.

4.5 Procedure

4.5.1 Immerse the surgical instrument in the test solution (4.3) at room temperature for 6 min. Remove the instrument and wash it with distilled water or wipe it with wet cotton wool. Examine the instrument for evidence of deposition of copper. Except as specified under 4.5.1, there shall be no plating of copper on the instrument.

4.5.2 Slight deposit of copper on the periphery of the instrument, or at the soldered or brazed junctions or dulling of polished surfaces caused by copper sulphate solution shall be disregarded. Similarly, slight plating of copper in small parts of joints, ratchets and serrations shall be disregarded.

5 BOILING WATER TEST

5.1 Reagent

Distilled water conforming to IS 1070.

5.2 Apparatus

Glass or ceramic beaker or suitable corrosion resistant stainless steel vessel.

5.3 Preparation of Sample

Scrub the instrument using soap and warm water, rinse thoroughly in distilled water (5.1) and dry.

5.4 Procedure

Immerse the instrument in boiling distilled water (see 5.1) in the beaker or vessel for at least 30 min. Subsequently allow the instrument to cool for at least 1 h in the water used for the test. Remove the instrument from the water and leave it exposed to the air for 2 h.

5.5 Rub the instrument vigorously with a dry cloth and examine it for the presence of blemishes or other signs of corrosion. Any blemish not removed by vigorous hand rubbing with a cloth shall be considered as evidence of corrosion.

6 AUTOCLAVING TEST

6.1 This test shall be performed subsequent to the boiling water test.

6.2 Autoclave

A suitable autoclave of the non-vacuum type, with a minimum performance of 134°C and 200 kN/m² (2.04kgf/cm²).

6.3 Water for Autoclaving Test

It shall conform to IS 1070.

6.4 Procedure

After cooling the instrument to room temperature following the boiling water test, place it unwrapped in a tray of the autoclave. Subject the instrument to six separate cycles of autoclaving each for $3 \pm \frac{1}{2}$ minutes at a minimum temperature and pressure of 134°C and 200 kN/m² (2.04 kgf/cm²) respectively. After each cycle, open the autoclave, remove the instrument and tray and allow it to cool to room temperature.

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6.5 Wipe the instrument with a dry cloth and inspect for any visible signs of corrosion. Any blemish not removed by vigorous hand rubbing with a cloth shall be considered as an evidence of corrosion.